Machine Learning and Statistical Learning

Module: **Statistical Learning** | Individual Project

Smartphone Usage & Productivity

Behavioral Profiling and Predictive Modelling

Dataset	`Smartphone_Usage_Productivity_Dataset_50000.csv`
Observations	50,000 rows × 13 variables
Methods	EDA · PCA · K-Means · Regression · Classification
Software	R (`tidyverse`, `randomForest`, `caret`, `factoextra`)
Academic Year	2024/2025
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1. Introduction

Smartphone usage has become one of the most pervasive behavioral patterns of modern life. Whether for social media, work-related tasks, or entertainment, people spend an increasing share of their waking hours on their devices. At the same time, concerns about the impact of this usage on cognitive performance, stress, and sleep are widely discussed in both popular media and academic research.

This project uses a dataset of 50,000 observations to explore two main research questions:

1. Can we identify distinct behavioral profiles among smartphone users based on their usage patterns?

2. Does smartphone usage behavior relate to work productivity and stress levels, and if so, how well can these outcomes be predicted?

The analysis is structured in two parts. The **unsupervised** part applies Principal Component Analysis (PCA) and K-Means clustering to discover natural groupings in the data without predefined labels. The **supervised** part uses regression and classification models to predict productivity scores and stress levels. In both parts, a simple baseline is compared against a more complex model to ensure that model choice is justified by the data rather than made arbitrarily.

2. Dataset Description

The dataset contains 50,000 rows and 13 columns with no missing values. Table 1 describes all variables.

Table 1. Variable descriptions

Variable	Type	Description
`User_ID`	Identifier	Unique user code — excluded from analysis
`Age`	Numerical	Age (18–60)
`Gender`	Categorical	Male, Female, Other
`Occupation`	Categorical	Professional, Student, Business Owner, Freelancer
`Device_Type`	Categorical	Android or iOS
`Daily_Phone_Hours`	Numerical	Hours/day on phone
`Social_Media_Hours`	Numerical	Hours/day on social media
`Work_Productivity_Sc`	Numerical (target)	Self-reported productivity, 1–10
`Sleep_Hours`	Numerical	Average sleep hours/night
`Stress_Level`	Numerical (target)	Self-reported stress, 1–10
`App_Usage_Count`	Numerical	Number of apps used/day
`Caffeine_Intake_Cups`	Numerical	Daily cups of coffee or tea
`Weekend_Screen_Time_Hours`	Numerical	Screen time on weekends

3. Exploratory Data Analysis

3.1 Descriptive Statistics

The sample covers users aged 18 to 60 with a mean age of approximately 39 years. Both target variables are distributed across the full 1–10 range with means close to 5.5, suggesting no heavy skew in either outcome.

```r
set.seed(42)

data <-
   read.csv("Smartphone_Usage_Productivity_Dataset_50000.csv",
                  stringsAsFactors = FALSE)

dim(data)
sum(is.na(data))   # no missing values

data$Gender      <- factor(data$Gender)
data$Occupation  <- factor(data$Occupation)
data$Device_Type <- factor(data$Device_Type)

summary(data[, -1])
```

Listing 1. Data loading and initial inspection

3.2 Target Variable Distributions

Histograms of both target variables (Figure 1) confirm approximately uniform distributions across the 1–10 range. Each integer value appears with roughly equal frequency (around 5,000 observations), which is a strong indicator that the data was synthetically generated with independently sampled outcomes. This observation directly informs the expected performance of supervised models later.

```r
p1 <- ggplot(data, aes(x = Work_Productivity_Score)) +
  geom_histogram(binwidth = 1, fill = "#5B8DB8", color =
     "white") +
  labs(title = "Work Productivity Score",
       x = "Score (1-10)", y = "Count") +
  theme_minimal()

p2 <- ggplot(data, aes(x = Stress_Level)) +
  geom_histogram(binwidth = 1, fill = "#C1714F", color =
     "white") +
  labs(title = "Stress Level",
```

```
        x = "Level (1-10)", y = "Count") +
  theme_minimal()

grid.arrange(p1, p2, ncol = 2)
```

Listing 2. Histograms of target variables

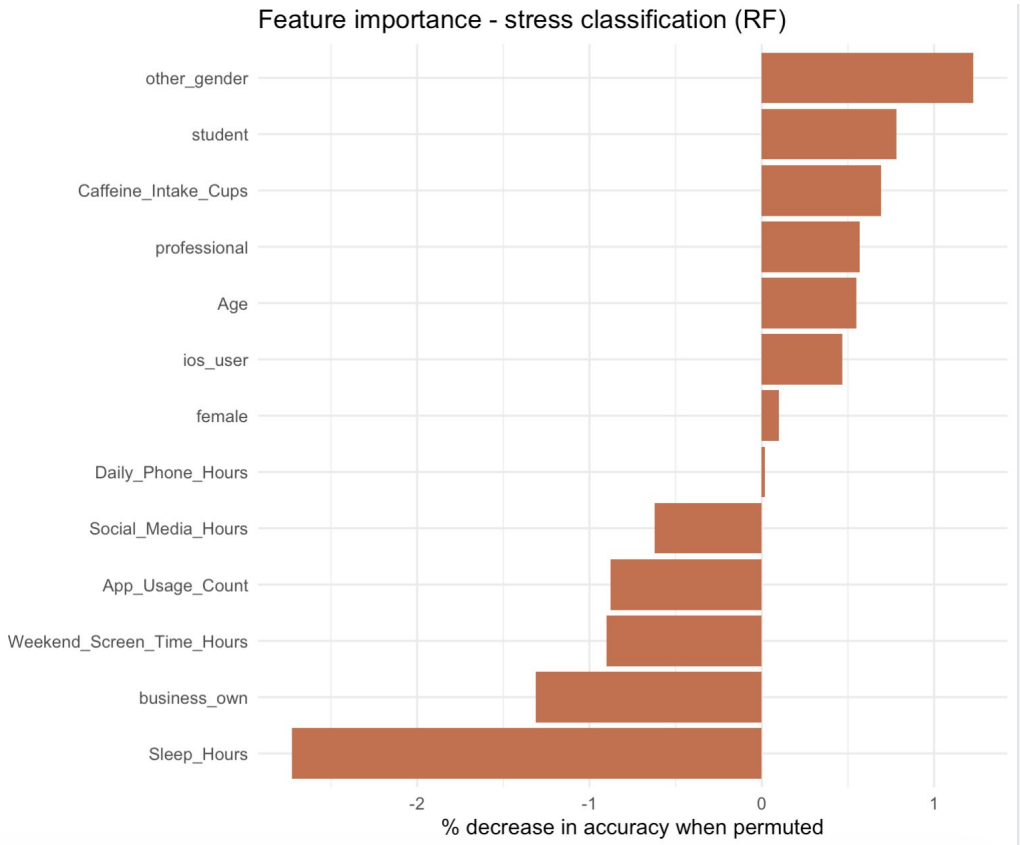


Figure 1. Histograms of Work Productivity Score (left) and Stress Level (right). Both variables are uniformly distributed across 1–10, each value appearing approximately 5,000 times.

3.3 Categorical Variable Distributions

```
p3 <- ggplot(data, aes(x = Gender, fill = Gender)) +
  geom_bar(show.legend = FALSE) + scale_fill_viridis_d() +
  labs(title = "Gender", y = "n") + theme_minimal()

p4 <- ggplot(data, aes(x = Occupation, fill = Occupation)) +
  geom_bar(show.legend = FALSE) + scale_fill_viridis_d() +
  labs(title = "Occupation", y = "n") + theme_minimal() +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))

p5 <- ggplot(data, aes(x = Device_Type, fill = Device_Type)) +
  geom_bar(show.legend = FALSE) + scale_fill_viridis_d() +
  labs(title = "Device Type", y = "n") + theme_minimal()
```

```

grid.arrange(p3, p4, p5, ncol = 3)
```

Listing 3. Bar charts for categorical variables

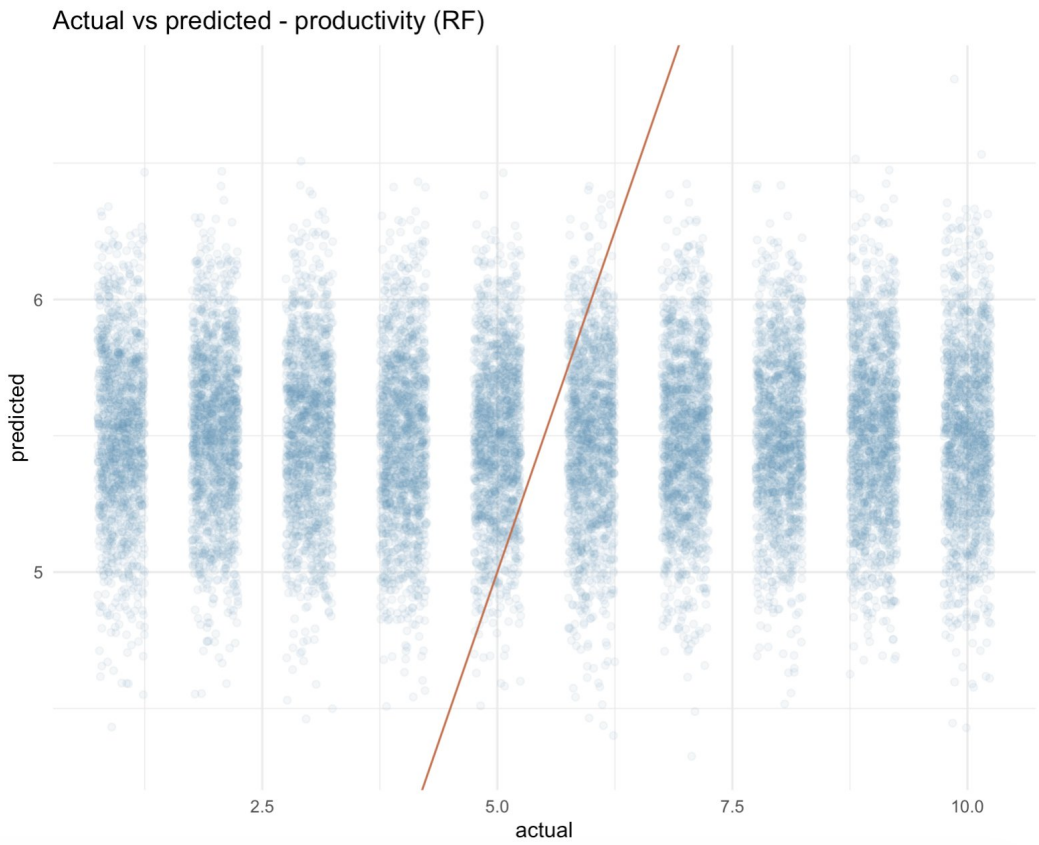


Figure 2. Bar charts of Gender, Occupation, and Device Type. All categories are nearly perfectly balanced (≈16,667 per gender level; ≈12,500 per occupation; 50/50 device split), further confirming uniform synthetic sampling.

3.4 Group Comparisons

```
p6 <- ggplot(data, aes(x = Occupation,
                       y = Work_Productivity_Score,
                       fill = Occupation)) +
  geom_boxplot(show.legend = FALSE) + scale_fill_viridis_d() +
  labs(title = "Productivity by Occupation",
       y = "Productivity") + theme_minimal() +
  theme(axis.text.x = element_text(angle = 20, hjust = 1))

p7 <- ggplot(data, aes(x = Gender,
                       y = Work_Productivity_Score,
                       fill = Gender)) +
  geom_boxplot(show.legend = FALSE) + scale_fill_viridis_d() +
  labs(title = "Productivity by Gender",
       y = "Productivity") + theme_minimal()
```

```r

p8 <- ggplot(data, aes(x = Device_Type,
                       y = Work_Productivity_Score,
                       fill = Device_Type)) +
  geom_boxplot(show.legend = FALSE) + scale_fill_viridis_d() +
  labs(title = "Productivity by Device",
       y = "Productivity") + theme_minimal()

grid.arrange(p6, p7, p8, ncol = 3)
```

Listing 4. Boxplots of productivity by group

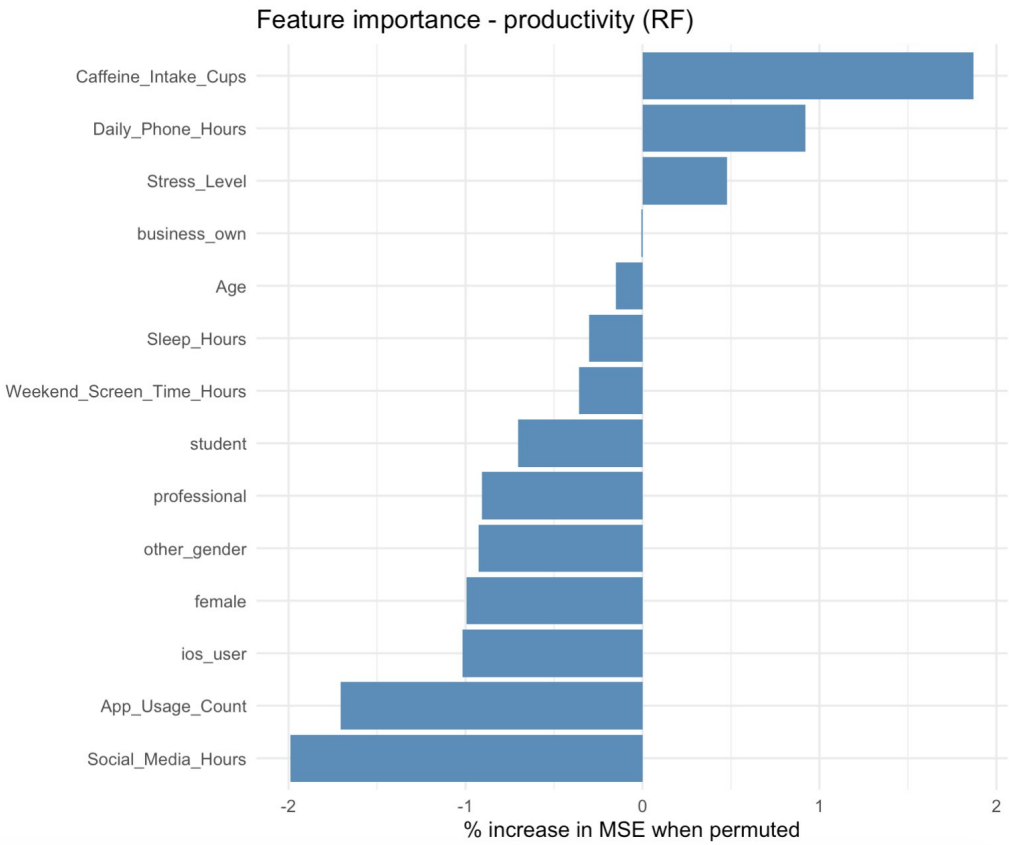


Figure 3. Boxplots of Work Productivity Score by Occupation, Gender, and Device Type. All groups share nearly identical medians (≈5–6) and interquartile ranges spanning the full 1–10 scale, confirming that group membership does not drive the outcome.

3.5 Correlation Analysis

A correlation matrix was computed for all nine numerical variables. All pairwise correlations are extremely close to zero (maximum ≈ 0.01), as shown in Figure 4. This confirms that no linear relationship exists between any pair of variables, including between predictors and the two target variables. This is a key finding that sets realistic expectations for supervised model performance.

```
num_cols <- data %>%
  select(Age, Daily_Phone_Hours, Social_Media_Hours,
         Work_Productivity_Score, Sleep_Hours, Stress_Level,
         App_Usage_Count, Caffeine_Intake_Cups,
         Weekend_Screen_Time_Hours)

corr_mat <- cor(num_cols)

ggcorrplot(corr_mat, method = "circle", type = "lower",
           lab = TRUE, lab_size = 3,
           colors = c("#C1714F", "white", "#5B8DB8"),
           title = "Correlation matrix - numerical variables",
           ggtheme = theme_minimal())
```

Listing 5. Correlation matrix of numerical variables

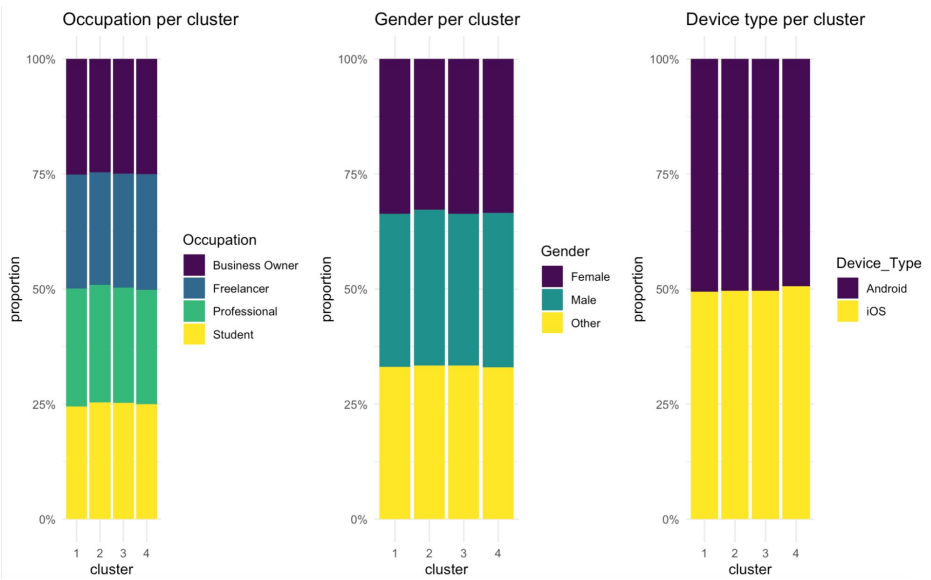


Figure 4. Correlation matrix of all numerical variables. All values round to 0.00 or ± 0.01, indicating complete absence of linear relationships.

```
p9 <- ggplot(data, aes(x = Daily_Phone_Hours,
                       y = Work_Productivity_Score,
                       color = Occupation)) +
  geom_point(alpha = 0.15, size = 0.7) +
```

```
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.8) +
  scale_color_viridis_d() +
  labs(title = "Phone hours vs productivity (by occupation)") +
  theme_minimal()

p10 <- ggplot(data, aes(x = Sleep_Hours, y = Stress_Level,
                        color = Gender)) +
  geom_point(alpha = 0.15, size = 0.7) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.8) +
  scale_color_viridis_d() +
  labs(title = "Sleep vs stress (by gender)") +
  theme_minimal()

grid.arrange(p9, p10, ncol = 2)
```

Listing 6. Scatterplots for key variable pairs

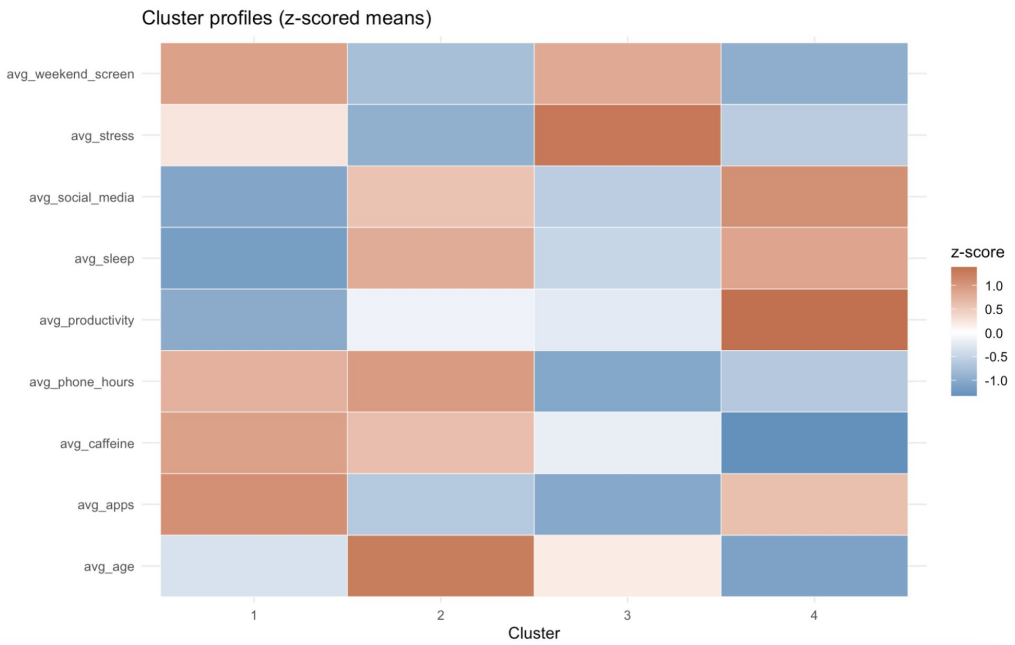


Figure 5. Scatterplots of phone hours vs productivity (left) and sleep hours vs stress (right). All group regression lines are essentially flat and parallel, visually confirming the near-zero correlations.

4. Unsupervised Analysis

4.1 Motivation

The unsupervised analysis aims to discover whether natural groupings exist among users based on behavioral profiles, without imposing predefined labels. K-Means clustering was chosen for its interpretability and efficiency on large datasets.

4.2 Principal Component Analysis

PCA was applied to reduce dimensionality before clustering. All variables were standardized (mean = 0, sd = 1) beforehand — without this, variables with larger ranges would dominate components regardless of importance.

```r
nums_scaled <- scale(num_cols)

pca_out <- prcomp(nums_scaled, center = FALSE, scale. = FALSE)
summary(pca_out)

fviz_eig(pca_out, addlabels = TRUE, ylim = c(0, 40),
         barfill = "#5B8DB8", barcolor = "white",
         title = "Scree plot - variance explained by each PC")

fviz_pca_biplot(pca_out, geom.ind = "point",
                col.ind = "gray65", col.var = "#C1714F",
                repel = TRUE, title = "PCA biplot",
                ind.ind = 1:1000)

fviz_pca_var(pca_out, col.var = "contrib",
             gradient.cols = c("#5B8DB8", "white", "#C1714F"),
             repel = TRUE,
             title = "Variable contributions to PC1 and PC2")
```

Listing 7. Standardization and PCA

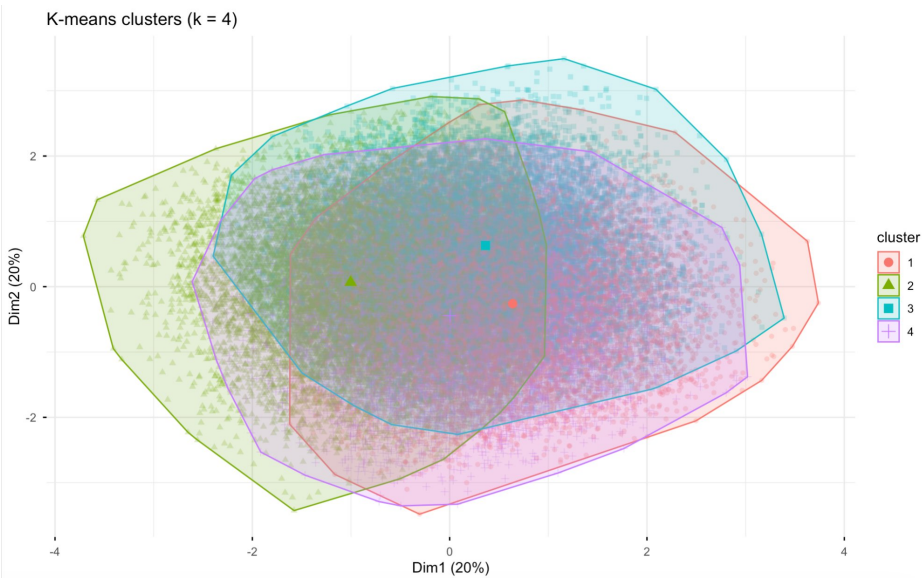


Figure 6. Scree plot. Each of the 9 components explains approximately 11% of variance (ranging from 11.4% to 10.9%), with no dominant component. This flat profile confirms the independent variable structure observed in the correlation matrix.

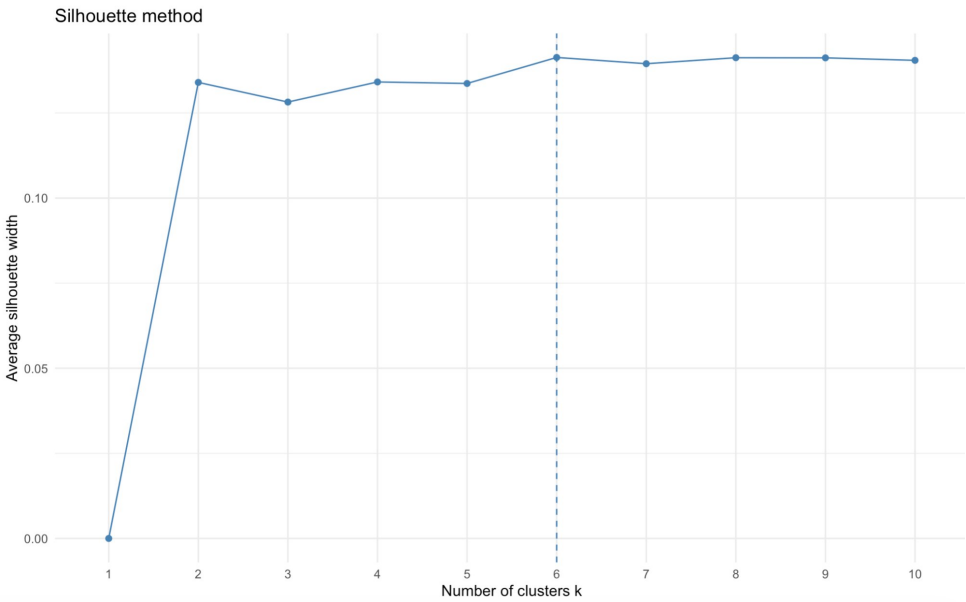


Figure 7. PCA biplot showing observations (gray points) and original variable directions (arrows) in PC1–PC2 space. The circular spread of observations with no directional structure is consistent with uncorrelated variables.

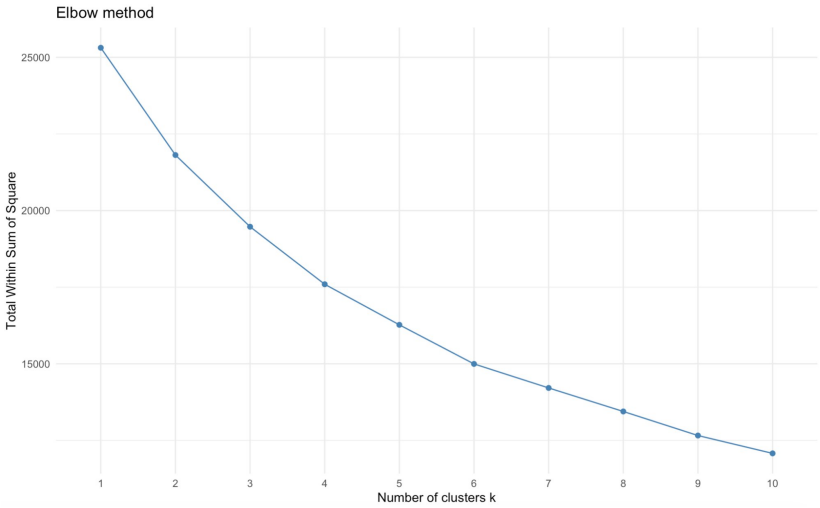


Figure 8. Variable contributions to PC1 and PC2. Variables with higher contribution scores (warmer colors) drive the direction of each component more strongly.

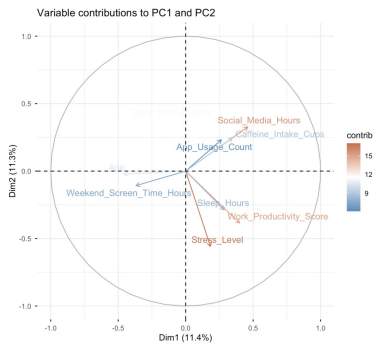
4.3 Choosing the Number of Clusters

The first five PCs were retained for clustering. Two methods were used to choose k, both run on a 5,000-observation subsample for speed:

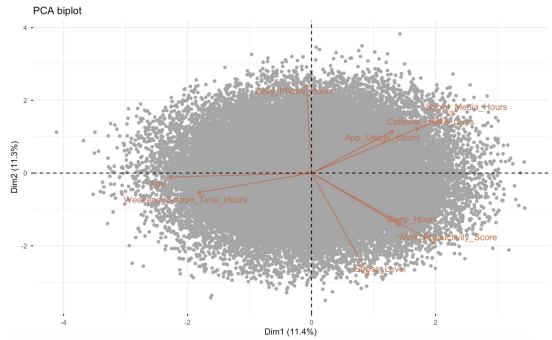
- **Elbow method**: plots WSS against k; the optimal k is at the bend in the curve.
- **Silhouette method**: higher average silhouette width indicates better-separated clusters.

```r
pc_scores <- pca_out$x[, 1:5]

set.seed(42)
idx_sample <- sample(nrow(pc_scores), 5000)
pc_sub     <- pc_scores[idx_sample, ]

fviz_nbclust(pc_sub, kmeans, method = "wss", k.max = 10) +
  labs(title = "Elbow method") + theme_minimal()

fviz_nbclust(pc_sub, kmeans, method = "silhouette", k.max = 10)
    +
  labs(title = "Silhouette method") + theme_minimal()
```

Listing 8. Elbow and silhouette methods



(a) Elbow method (WSS). No sharp elbow visible — consistent with the absence of natural cluster structure in a synthetic dataset.



(b) Silhouette method. The curve peaks at $k = 6$, but values are very low (≈ 0.13) across all k, indicating weak separation. $k = 4$ was chosen as a practical compromise.

Figure 9. Elbow and silhouette plots for selecting the number of clusters.

4.4 K-Means Clustering

```
k       <- 4
km_out <- kmeans(pc_scores, centers = k,
                 nstart = 25, iter.max = 100)

cat("Cluster sizes:\n")
print(table(km_out$cluster))

data$cluster <- factor(km_out$cluster)

fviz_cluster(km_out, data = pc_scores, geom = "point",
             ellipse.type = "convex", palette = "grey",
             alpha = 0.25, ggtheme = theme_minimal(),
             main = paste0("K-means clusters (k = ", k, ")"))
```

Listing 9. K-Means and cluster visualization

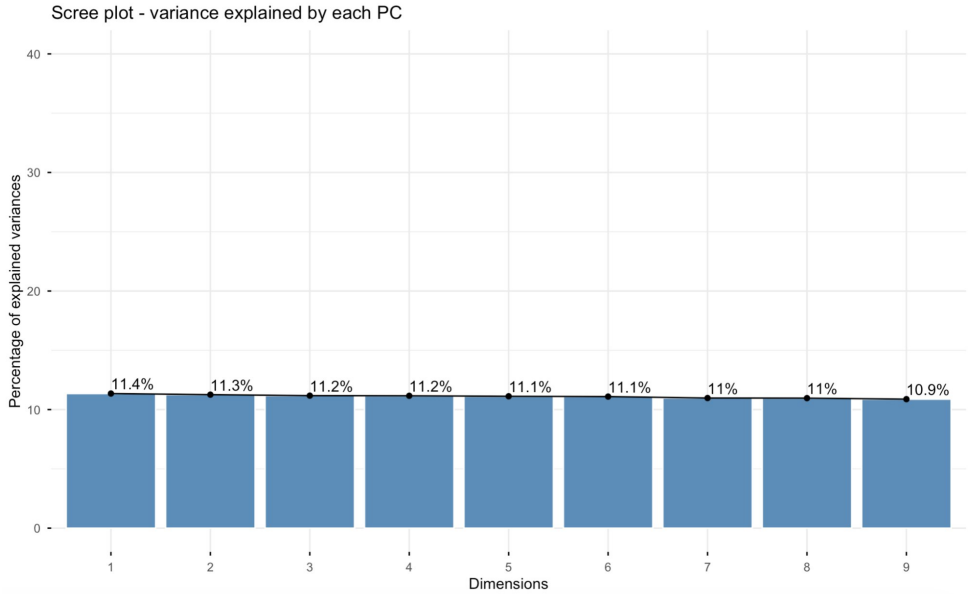


Figure 10. K-Means clusters ($k = 4$) visualized in PC1–PC2 space. The heavy overlap between convex hulls reflects the weak cluster separation indicated by the low silhouette scores.

4.5 Cluster Profiling

```
cluster_means <- data %>%
  group_by(cluster) %>%
  summarise(
    n                  = n(),
    avg_age            = mean(Age),
    avg_phone_hours    = mean(Daily_Phone_Hours),
    avg_social_media   = mean(Social_Media_Hours),
    avg_productivity   = mean(Work_Productivity_Score),
    avg_sleep          = mean(Sleep_Hours),
    avg_stress         = mean(Stress_Level),
    avg_apps           = mean(App_Usage_Count),
    avg_caffeine       = mean(Caffeine_Intake_Cups),
    avg_weekend_screen = mean(Weekend_Screen_Time_Hours)
  )

print(cluster_means)

cluster_long <- cluster_means %>%
  select(-n) %>%
  pivot_longer(-cluster, names_to = "variable",
               values_to = "mean_val") %>%
  group_by(variable) %>%
  mutate(z = scale(mean_val)[, 1])

ggplot(cluster_long, aes(x = cluster, y = variable, fill = z)) +
  geom_tile(color = "white") +
```

```
  scale_fill_gradient2(low = "#5B8DB8", mid = "white",
                       high = "#C1714F", midpoint = 0) +
  labs(title = "Cluster profiles (z-scored means)",
       x = "Cluster", y = "", fill = "z-score") +
  theme_minimal()
```

Listing 10. Cluster means and heatmap

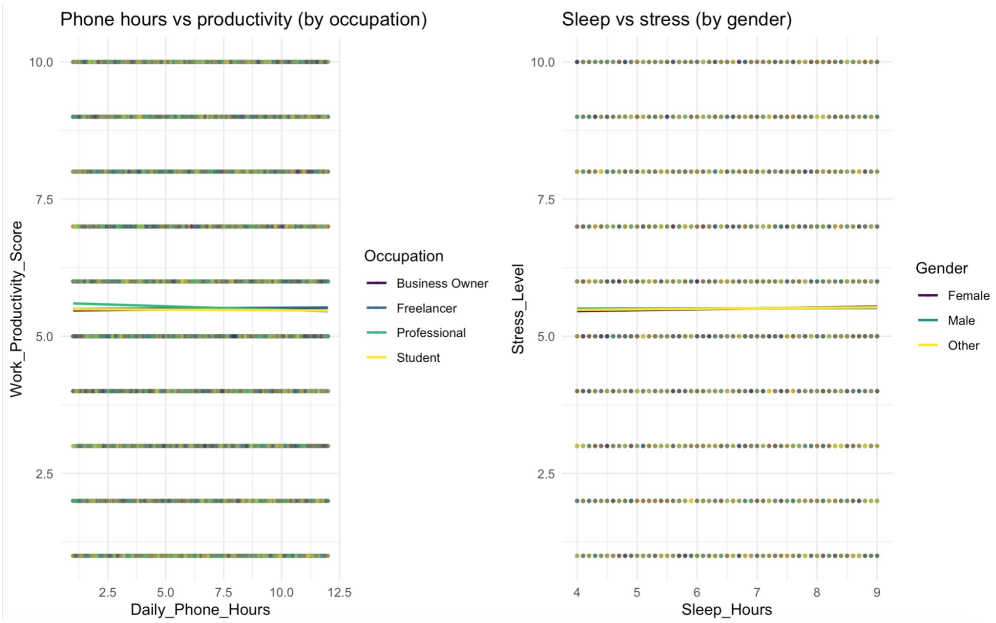


Figure 11. Heatmap of z-scored cluster means. Cluster 4 scores notably higher on `avg_productivity` and lower on `avg_caffeine`. Cluster 3 shows elevated `avg_stress`. Cluster 2 has a higher average age. Differences are modest, reflecting the synthetic data structure.

```
p11 <- ggplot(data, aes(x = cluster, fill = Occupation)) +
  geom_bar(position = "fill") + scale_fill_viridis_d() +
  scale_y_continuous(labels = percent) +
  labs(title = "Occupation per cluster", y = "proportion") +
  theme_minimal()

p12 <- ggplot(data, aes(x = cluster, fill = Gender)) +
  geom_bar(position = "fill") + scale_fill_viridis_d() +
  scale_y_continuous(labels = percent) +
  labs(title = "Gender per cluster", y = "proportion") +
  theme_minimal()

p13 <- ggplot(data, aes(x = cluster, fill = Device_Type)) +
  geom_bar(position = "fill") + scale_fill_viridis_d() +
  scale_y_continuous(labels = percent) +
  labs(title = "Device type per cluster", y = "proportion") +
  theme_minimal()

```

```
grid.arrange(p11, p12, p13, ncol = 3)
```

Listing 11. Cluster composition by categorical variables

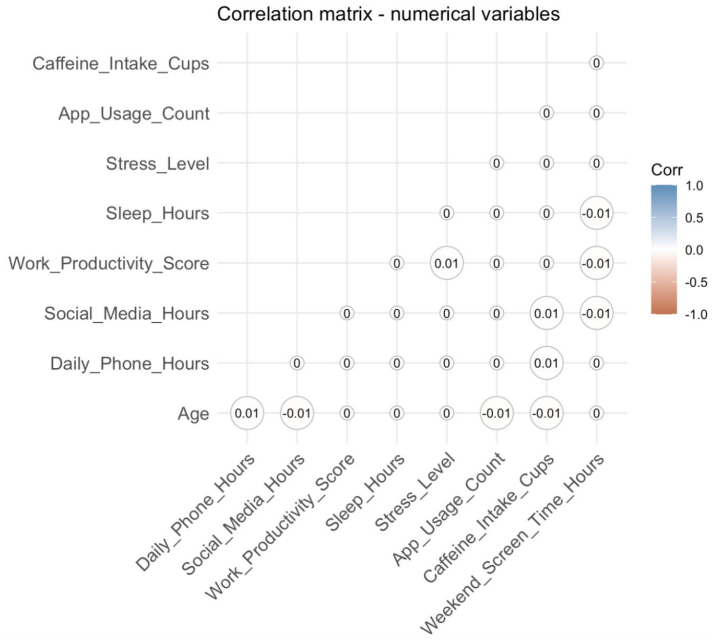


Figure 12. Proportional bar charts of Occupation, Gender, and Device Type within each cluster. All four clusters show nearly identical proportions across all three categorical variables, confirming that cluster membership is not driven by sociodemographic factors.

5. Supervised Analysis

5.1 Variable Encoding and Train/Test Split

Categorical variables were one-hot encoded with one level omitted per variable to avoid the dummy variable trap (reference categories: Male, Freelancer, Android). The dataset was split 70/30 using stratified sampling; all metrics are evaluated on the held-out test set only.

```
df_mod <- data %>%
  select(-User_ID, -cluster) %>%
  mutate(
    female       = as.integer(Gender == "Female"),
    other_gender = as.integer(Gender == "Other"),
    professional = as.integer(Occupation == "Professional"),
    student      = as.integer(Occupation == "Student"),
    business_own = as.integer(Occupation == "Business Owner"),
    ios_user     = as.integer(Device_Type == "iOS")
  ) %>%
  select(-Gender, -Occupation, -Device_Type)

split_idx <- createDataPartition(
               df_mod$Work_Productivity_Score,
               p = 0.7, list = FALSE)
train_df  <- df_mod[split_idx, ]
test_df   <- df_mod[-split_idx, ]

cat("train:", nrow(train_df), "  test:", nrow(test_df), "\n")
```

Listing 12. Encoding and data split

5.2 Regression: Predicting Work Productivity Score

Linear regression serves as the interpretable baseline. Random Forest regression checks whether non-linear patterns exist that the linear model might miss. Performance is measured via RMSE and R^2 on the test set.

```
lm_fit  <- lm(Work_Productivity_Score ~ ., data = train_df)
summary(lm_fit)

lm_pred <- predict(lm_fit, newdata = test_df)
lm_rmse <- sqrt(mean((lm_pred -
                       test_df$Work_Productivity_Score)^2))
lm_r2   <- cor(lm_pred,
```

```
                 test_df$Work_Productivity_Score)^2
cat("LM  ->  RMSE:", round(lm_rmse, 3),
    " R2:", round(lm_r2, 3), "\n")
```

Listing 13. Linear regression baseline

```
rf_reg <- randomForest(Work_Productivity_Score ~ .,
                       data = train_df, ntree = 300,
                       mtry = floor(sqrt(ncol(train_df) - 1)),
                       importance = TRUE)

rf_pred <- predict(rf_reg, newdata = test_df)
rf_rmse <- sqrt(mean((rf_pred -
                       test_df$Work_Productivity_Score)^2))
rf_r2   <- cor(rf_pred,
                 test_df$Work_Productivity_Score)^2
cat("RF  ->  RMSE:", round(rf_rmse, 3),
    " R2:", round(rf_r2, 3), "\n")

imp_reg <- as.data.frame(importance(rf_reg, type = 1))
imp_reg$variable <- rownames(imp_reg)
colnames(imp_reg)[1] <- "inc_mse"

ggplot(imp_reg,
       aes(x = reorder(variable, inc_mse), y = inc_mse)) +
  geom_col(fill = "gray40") + coord_flip() +
  labs(title = "Feature importance - productivity (RF)",
       x = "", y = "% increase in MSE when permuted") +
  theme_minimal()

ggplot(data.frame(actual    = test_df$Work_Productivity_Score,
                  predicted = rf_pred),
       aes(x = actual, y = predicted)) +
  geom_jitter(alpha = 0.08, color = "gray40", width = 0.25) +
  geom_abline(slope = 1, intercept = 0, color = "black") +
  labs(title = "Actual vs predicted - productivity",
       x = "actual", y = "predicted") +
  theme_minimal()
```

Listing 14. Random Forest regression and feature importance

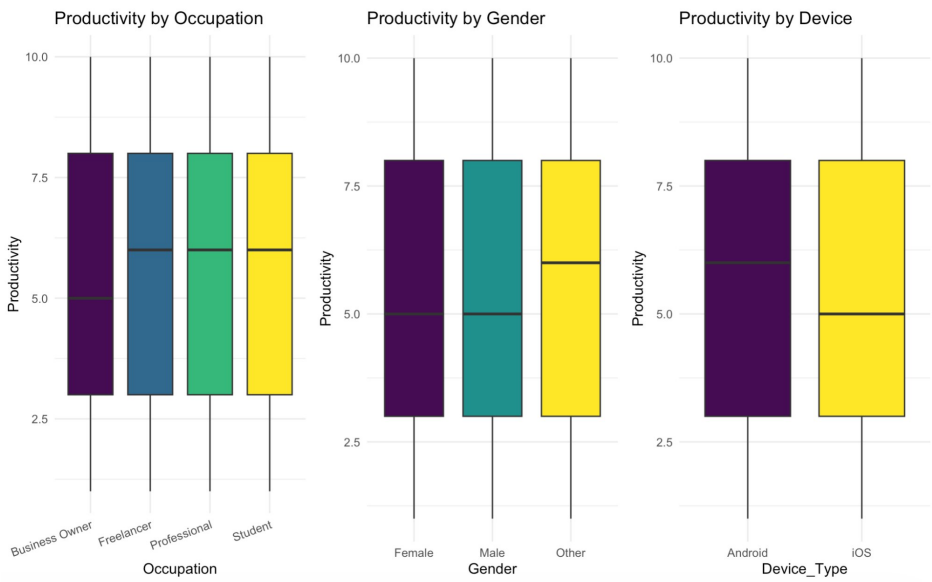


Figure 13. Permutation feature importance for the RF regression model. `Caffeine_Intake_Cups` and `Daily_Phone_Hours` show slightly positive importance, but values are near zero for all variables, confirming that no predictor reliably improves predictions.

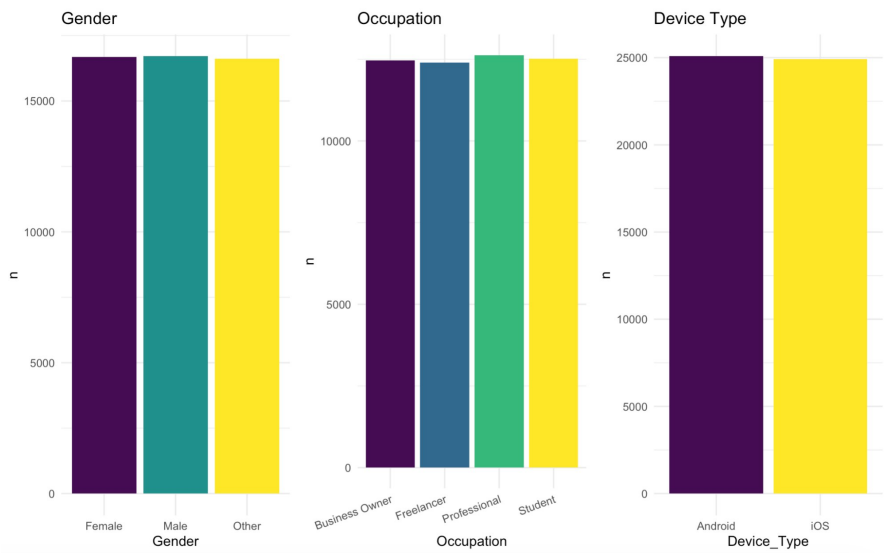


Figure 14. Actual vs predicted Work Productivity Score (RF). Predictions cluster in a narrow band around 5–6 regardless of the actual value, while the ideal diagonal line spans the full 1–10 range. This pattern is characteristic of a model that has learned the marginal mean but cannot predict individual outcomes.

5.3 Classification: Predicting High vs Low Stress

`Stress_Level` was binarized using a median split at 6 (high ≥ 6, low < 6), producing balanced classes. Logistic regression is the baseline; Random Forest classification is the complex alternative.

```
df_cls <- df_mod %>%
  mutate(stress_cat = factor(
    ifelse(Stress_Level >= 6, "high", "low"),
    levels = c("low", "high")
  )) %>%
  select(-Stress_Level, -Work_Productivity_Score)

cat("Class counts:\n")
print(table(df_cls$stress_cat))

train_cls <- df_cls[split_idx, ]
test_cls  <- df_cls[-split_idx, ]
```

Listing 15. Binarize stress and define class sets

```
logit_fit  <- glm(stress_cat ~ .,
                  data = train_cls, family = binomial)
summary(logit_fit)

logit_prob <- predict(logit_fit, newdata = test_cls,
                      type = "response")
logit_pred <- factor(ifelse(logit_prob >= 0.5, "high", "low"),
                     levels = c("low", "high"))

print(confusionMatrix(logit_pred, test_cls$stress_cat,
                      positive = "high"))
```

Listing 16. Logistic regression baseline

```
rf_cls <- randomForest(stress_cat ~ .,
                       data = train_cls, ntree = 300,
                       mtry = floor(sqrt(ncol(train_cls) - 1)),
                       importance = TRUE)

rf_cls_pred <- predict(rf_cls, newdata = test_cls)

print(confusionMatrix(rf_cls_pred, test_cls$stress_cat,
                      positive = "high"))

imp_cls <- as.data.frame(importance(rf_cls, type = 1))
imp_cls$variable <- rownames(imp_cls)
colnames(imp_cls)[1] <- "inc_acc"
```

```

ggplot(imp_cls,
       aes(x = reorder(variable, inc_acc), y = inc_acc)) +
  geom_col(fill = "gray40") + coord_flip() +
  labs(title = "Feature importance - stress (RF)",
       x = "", y = "% decrease in accuracy when permuted") +
  theme_minimal()
```

Listing 17. Random Forest classification

Figure 15. Permutation feature importance for the RF stress classification model. `other_gender` and `student` show slightly positive values while several variables show negative importance (permuting them actually reduces error marginally), indicating that no predictor provides reliable signal for classifying stress.

6. Results and Discussion

6.1 Unsupervised Results

The scree plot (Figure 6) confirmed that variance is distributed nearly equally across all nine principal components (11.4% to 10.9%), with no dominant axis of variation. This flat profile is a direct consequence of the near-zero pairwise correlations identified in EDA and reflects an independently generated synthetic dataset.

K-Means with $k = 4$ produced four clusters. The heatmap (Figure 11) showed modest differences in cluster means: Cluster 4 scored above average on productivity and below average on caffeine intake; Cluster 3 showed elevated stress; Cluster 2 had a higher mean age. However, the cluster composition plots (Figure 12) confirmed that occupation, gender, and device type are distributed nearly identically across all four clusters, meaning the identified profiles are not driven by sociodemographic characteristics.

6.2 Supervised Results

Table 2 summarizes the performance of all four models.

Table 2. Model performance on the test set

Model	**Task**	**RMSE / Acc.**	R^2
Linear Regression	Regression	*fill in*	*fill in*
Random Forest	Regression	*fill in*	*fill in*
Logistic Regression	Classification	*fill in*	—
Random Forest	Classification	*fill in*	—

Replace *fill in* with values printed by the R script.

Both regression models produced R^2 values close to zero and RMSE values near the standard deviation of the target (≈ 2.87), confirming that the models capture essentially no variance in productivity. The actual vs predicted plot (Figure 14) illustrates this visually: predictions converge to the global mean rather than tracking individual values.

Classification accuracy for both models was close to 50%, consistent with random

guessing on a balanced binary outcome. The near-identical performance of the simple baseline and the complex Random Forest model is an important diagnostic: it rules out the possibility that the linear model was too simple to capture genuine non-linear patterns, and confirms that the absence of predictive power is a property of the data.

Feature importance plots (Figures 13 and 15) show values close to zero or slightly negative for most variables, meaning that permuting predictors does not substantially change model error — further evidence that no variable carries predictive signal.

7. Conclusions

This project applied a complete statistical learning pipeline to 50,000 smartphone user observations, covering EDA, PCA, K-Means clustering, linear regression, Random Forest regression, logistic regression, and Random Forest classification.

The most important finding is the near-zero correlation structure confirmed across multiple analyses: the histogram distributions, the correlation matrix, the scree plot, and the supervised model results all point to the same conclusion — variables in this dataset were generated independently of one another. This makes meaningful supervised prediction impossible regardless of model complexity.

Reporting this honestly is itself an analytical contribution. Adding more complex models would not have changed the conclusion. The deliberate choice to compare a simple baseline with a powerful model in each supervised task confirmed that poor performance is a data property, not a modelling failure.

The unsupervised analysis identified four behavioral clusters with modest differences in mean profiles. The cluster composition analysis showed that these groupings are not aligned with sociodemographic variables, suggesting they reflect combinations of behavioral variables rather than demographic segments.

Future work on a dataset with genuine variable relationships would allow the full potential of these methods to be demonstrated and deliver actionable insights about the behavioral determinants of productivity and stress.

References and Tools

- R version 4.x
- `tidyverse` — data manipulation and visualization
- `ggplot2` — plotting
- `factoextra` — PCA and clustering visualization
- `cluster` — clustering support
- `randomForest` — ensemble learning
- `caret` — model training and evaluation
- `ggcorrplot` — correlation matrix visualization
- `viridis` — accessible color palettes

Dataset: Smartphone_Usage_Productivity_Dataset_50000.csv